EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

1.Fact-finding Techniques

To identify authentic and detailed requirements, our team used two complementary fact-finding methods: **one-on-one user interviews** and **structured questionnaires**. This combination leverages their synergies—interviews explore personalized insights from experienced users, while questionnaires quantify common needs across a broader user base.

a. One-on-One User Interviews

Rationale

This method uncovers **in-depth, personalized insights not captured by closed-ended questionnaires**. We interviewed a senior student with TA experience to gather first-hand feedback on the current manual (form/Excel) recruitment process and expectations for the new system, forming the basis for core user stories.

Implementation

Key Findings

1. **Core Pain Points**: Repeated data entry, no application status visibility, scattered job postings, unclear job details causing schedule conflicts, and inefficient email-only communication with MOs.
2. **Must-Have Features**: Real-time application status inquiry (US-06), one-click CV upload (US-02), and a centralized job browsing page (US-03).
3. **Usability Needs**: Job filtering/search (US-07), dedicated MO communication channel (US-08), and a clear, intuitive interface.

b. Structured Questionnaire

Rationale

Questionnaires **quantify common needs and preferences** across potential TA applicants,

verifying the universality of interview findings and informing feature prioritization. This ensures our High/Medium/Low priority ranking reflects the majority user needs.

Implementation

Key Findings

1. **Universal Pain Points**: Scattered job postings, and no status visibility were top concerns, elevating US-03 and US-06 to **High priority**.
2. **Critical Job Details**: Required skills, workload, and deadlines were the most important to applicants, guiding mandatory fields in US-11 (Post a Job).
3. **Function Demand**:
    - 80% needed job filtering (US-07, Medium priority);
    - Some wanted AI job recommendation (US-09) and skill feedback (US-10, Low priority for later iterations);
    - Online MO communication (US-08) was deemed essential (Medium priority).
4. **System Rule**: Support schedule conflict checks (US-15).

c. Impact on Requirements and Prioritization

We prioritized features by **suggestion frequency, implementation difficulty, and functional fundamentality**:

- **High priority**: Core pain-point solving functions;
- **Medium priority**: Usability improvements;
- **Low priority**: Complex AI features for later iterations.

Detailed requirements (e.g., mandatory fields, notification methods) were embedded into the **acceptance criteria** of corresponding user stories, ensuring a practical, user-centric design.

2. Iteration Plan

The project includes an initial planning stage followed by four development Sprints. According to the module handout, the First Assessment on 22 March 2026 requires the submission of the product backlog, prototype, and brief report rather than a completed software release. Therefore, fact-finding, story writing, backlog preparation, and prototyping are completed before 22 March 2026, and implementation begins immediately afterwards.

Sprint	User Stories	Period
Sprint 1	US-00, US-01, US-02, US-03, US-04, US-11	23 Mar – 2 Apr 2026
Sprint 2	US-05, US-06, US-12, US-13, US-14	6 Apr – 17 Apr 2026
Sprint 3	US-07, US-08, US-15	20 Apr – 2 May 2026
Sprint 4	US-09, US-10	4 May – 15 May 2026

Sprint 1 delivers the foundational end-to-end recruitment flow. User registration and login (US-00) is implemented first, as it is a prerequisite for all role-specific functionality. Once authentication is in place, the remaining Sprint 1 stories are developed in parallel: a TA

applicant can create a profile (US-01), submit a CV (US-02), browse open jobs (US-03), and apply for a position (US-04), while a Module Organiser can post a job (US-11). This Sprint aims to produce Working Software Version 1 by 29 March 2026.

Sprint 2 extends the system with status tracking, job management, and administrative controls. TAs can view their application status (US-06) and edit their profiles (US-05). Module Organisers can review applications and update candidate status (US-13), as well as edit or close job postings (US-12). The Admin workload dashboard (US-14) is also delivered in this Sprint, providing the schedule and hours data that US-15 depends on. Sprint 2 spans the Intermediate Assessment on 12 April 2026, and a demonstrable version of the software will be available at that point, with the remaining Sprint 2 tasks completed shortly afterwards.

Sprint 3 introduces usability improvements and the schedule conflict prevention feature. Job search and filtering (US-07) and applicant-to-MO messaging (US-08) enhance the user experience. Schedule conflict prevention (US-15) is placed in Sprint 3 rather than Sprint 2 because it depends on both the schedule data introduced in US-11 and the workload data computed by US-14, making earlier delivery technically infeasible.

Sprint 4 focuses on advanced and optional AI-assisted features: job recommendation based on skills (US-09) and missing skills feedback (US-10). These are deferred due to their low priority, high story point estimates, and dependency on a stable core system.

After Sprint 4, a final stabilisation period from 16 May to 24 May 2026 is reserved for testing, bug fixing, documentation, demonstration video preparation, and final submission. Each Sprint concludes with a team retrospective to adjust plans for the subsequent Sprint where necessary.

3. Prioritisation

User stories were prioritised using the **MoSCoW method**, as introduced in the module lectures. This approach classifies each story into one of four categories — **Must have**, **Should have**, **Could have**, and **Won't have** — based on business value, user need, and technical feasibility. Prioritisation decisions were also informed by findings from our fact-finding activities, including survey responses and user interviews.

Must have features are essential to the system's core operation; without them, the system would have no practical value. These are: user registration and login (US-00), creating an applicant profile (US-01), submitting a CV (US-02), browsing open jobs (US-03), applying for a job (US-04), viewing application status (US-06), posting a job (US-11), reviewing and processing applications (US-13), checking TA workload (US-14), and preventing schedule conflicts (US-15). These stories cover the complete recruitment workflow from both the TA and Module Organiser perspective, and were consistently identified as essential by participants in our fact-finding activities. Although US-15 is delivered in Sprint 3 due to its technical dependencies, its business importance warrants a Must have classification.

Should have features are important for usability and data integrity, but their absence would not prevent the core recruitment process from functioning. These are: editing an applicant

profile (US-05) and editing or closing a job posting (US-12). Both support ongoing system use beyond initial setup but can be deferred to Sprint 2 without breaking the primary workflow.

Could have features are desirable enhancements planned for delivery in Sprint 3 and beyond. These include: searching and filtering jobs (US-07), contacting a Module Organiser (US-08), and the two AI-assisted features — job recommendation based on skills (US-09) and missing skills feedback (US-10). Users acknowledged these as useful during fact-finding but did not consider them critical for early delivery. The AI-assisted stories (US-09, US-10) carry the highest story point estimates (8 and 5 respectively) and are deferred to Sprint 4, to be implemented only if the core system is fully stable and sufficient development time remains.

Won't have in the current release cycle are features entirely outside the defined project scope, such as third-party integrations or administrative reporting beyond workload tracking; no current backlog items fall into this category.

4. Estimation Method

To ensure reasonable iteration planning, the team adopted the **Story Points** estimation technique as required in the course materials, with the **Fibonacci sequence (1, 2, 3, 5, 8, 13)** serving as the scoring scale.

a. Method and Principle: Story Points + Fibonacci Sequence

Core of the Method

It replaces absolute time estimation (e.g., person-days) with "relative workload comparison", comprehensively considering **technical complexity, development workload, and risks**. This avoids estimation deviations caused by differences in team members' technical proficiency or changes in requirements, and aligns with the iterative characteristics of Agile development.

Reason for Choosing the Fibonacci Sequence

The non-linear scale of the Fibonacci sequence (e.g., 3→5) truly reflects the principle that "**the larger the task, the higher the uncertainty**":

- For simple tasks, the team can accurately judge the workload difference between small intervals;
- For complex tasks (such as AI integration), larger intervals (e.g., 5→8) are used to avoid overconfident "precise estimation" and reduce deviations.

b. Operation Process: Baseline Reference + Team Consensus

Confirming the Baseline Story

US-01 (Create Applicant Profile) was designated as the baseline, with a score of 3 Story Points. The reasons are:

- It is a core function with moderate complexity, involving form design, data validation, and storage;
- All team members have a unified understanding of the "3-point workload" standard.

Estimation Steps

1. The Product Owner (PO) communicates the requirements and acceptance criteria of each user story to the team;
2. Team members independently score each story based on the Fibonacci sequence;
3. When there are disagreements, the team discusses technical difficulties (e.g., application restrictions for US-04) and records the result in the Backlog after reaching a consensus.

c. Estimation Examples

Example 1: US-08 (Contact Module Organiser) — 5 Points

This function requires implementing online message interaction with Module Organisers, including message sending, receiving, read status marking, and historical message storage. It needs additional development of message status management logic. Compared with the baseline US-01, it has more interaction processes and data associations, resulting in higher complexity. Therefore, it is estimated at 5 points.

Example 2: US-09 (Job Recommendation Based on Skills) — 8 Points

It has high technical complexity: it requires connecting to AI interfaces, handling fault-tolerance logic (e.g., timeout/data anomalies), and repeated testing of the accuracy of AI results. Its uncertainty is much higher than that of US-01, so it is estimated at 8 points.

d. Result Guarantee

All estimations require the approval of "over 80% of team members". For high-priority stories (e.g., US-00), additional review meetings are held to confirm the workload of technical difficulties, providing a basis for task allocation in iterations.

2. Supporting material

Include supporting materials, such as interview questions, survey results, and customer feedback etc (if applicable).

1. Fact-Finding Overview

Two primary fact-finding techniques were used to gather requirements: a structured questionnaire distributed to potential TA applicants, and a semi-structured one-on-one interview with a graduating senior who had prior TA application and working experience. The findings from both activities directly informed the product backlog, user story acceptance criteria, and prioritisation decisions described in the main report.

2. Survey

Purpose: To collect quantitative data from a broader pool of potential TA applicants regarding their pain points with the current application process, their feature preferences, and their interface expectations.

Distribution: The survey was distributed digitally to students across multiple programmes at BUPT International School.

Key findings and their mapping to user stories:

Survey Finding	Related User Story
Majority of respondents cited scattered job information and the need to check multiple channels as their main pain point	US-03 (Browse Open Jobs), US-07 (Search and Filter Jobs)
Repeatedly filling in personal information for every application was a widely reported inconvenience	US-01 (Create Applicant Profile), US-02 (Submit CV)
Not knowing the status of submitted applications after submission was the most commonly cited frustration	US-06 (View Application Status)
Most respondents ranked workload and working hours as one of the top three pieces of information they need from a job posting	US-11 (Post a Job), US-14 (Check TA Workload)
Majority preferred in-system reminders for application updates, with email as a secondary backup	US-06 (View Application Status)
Respondents who expressed interest in automated job recommendations based on their CV	US-09 (Job Recommendation Based on Skills)
Willingness to upload CV and personal information to assist with job matching was high	US-01, US-02
Most preferred filtering by position type, working hours, and skill requirements when browsing jobs	US-07 (Search and Filter Jobs)
Most respondents prioritised a simple, clearly sectioned interface that works on both desktop and mobile	General system design

Note on survey scope: Several survey questions (Q6 on a job favourites/bookmarking feature, Q10 on displaying applicant competition levels, Q11 on one-click bulk application, and Q15 on making hired TA profiles publicly visible) were included for exploratory purposes during early fact-finding. After reviewing feasibility and alignment with the core project requirements, the features raised by these questions were evaluated and either

excluded from the current project scope or not prioritised in the backlog. This reflects a deliberate scoping decision rather than an oversight.

Survey Questionnaire (full text):

1. Do you have any on-campus/off-campus TA work experience?

A. No TA experience at all / B. 1 TA experience / C. 2–3 TA experiences / D. 4 or more TA experiences

2. In the past, if you applied for a TA position, what main channels did you use to get recruitment information? *(Multiple choices allowed)*

A. University official website / college notices / B. Class group / supervisor notices / C. Peer recommendations / D. Offline posters / E. Others

When applying for a TA position, what core information about the recruitment position do you care about the most? *(Multiple choices allowed, maximum 3 items)*

A. Course/activity type / B. Skills and qualifications required / C. Workload and working hours / D. Application deadline and result notification timeline / E. Other supporting benefits/requirements

3. If you have applied for a TA position before, what is the most inconvenient part of the current application method using forms/Excel? *(Multiple choices allowed)*

A. Scattered position information / B. Repeated entry of personal information / C. No visibility of follow-up review progress / D. No application experience / E. Others

4. What methods do you prefer to receive recruitment reminders? *(Multiple choices allowed)*

A. Real-time in-system reminders / B. Email / C. SMS / D. No reminders / E. Others

5. Would you like the system to automatically recommend suitable TA positions based on your resume?

A. Definitely yes / B. Doesn't matter / C. No, I prefer to search on my own

6. Are you willing to fill in and upload relevant information to assist with job matching?

A. Fully willing / B. Reluctantly accept / C. Refuse to upload

7. When searching for TA positions, what screening criteria do you need the most? *(Multiple choices allowed)*

A. Position type / B. Affiliated college / course / C. Working hours / D. Remuneration / E. Number of recruits / F. Application deadline / G. Position requirements

8. Do you think the online communication function (consult the person in charge about position-related questions) is necessary?

A. Absolutely necessary / B. Doesn't matter / C. No, I prefer email or other means

9. What requirements do you have for the operation interface?

A. Simple and easy to understand / B. Clear functional division / C. Aesthetically pleasing / D. Adaptable to computer/mobile terminals / E. Others

10. What do you think is the most core and indispensable function of the TA recruitment system? *(Open-ended)*
11. Do you have any additional functional suggestions or usage expectations? *(Open-ended)*

3. Interview

Topic: Investigating past TA applicants' experience with the campus TA recruitment process and their functional requirements and design suggestions for the TA recruitment system

Interviewee: Senior Xiao (graduating student with TA application and working experience)

Interviewer: Qixin Bi

Date: 14 March 2026

Format: One-on-one offline interview

Key insights and their mapping to user stories:

Interview Insight	Related User Story
Job information was scattered across multiple channels; applicants frequently missed openings due to lack of a centralised listing	US-03 (Browse Open Jobs)
Applicants had to repeatedly enter the same personal information for every application	US-01 (Create Applicant Profile), US-02 (Submit CV)
No visibility of application progress after submission; applicants were left without feedback	US-06 (View Application Status)
Job postings lacked detail on workload and working hours, causing schedule conflicts after hiring	US-11 (Post a Job), US-15 (Prevent Schedule Conflict)
Communicating with Module Organisers post-hiring was difficult; email was the only available channel and responses were slow	US-08 (Contact Module Organiser)
Filtering by position type and working hours when browsing jobs was described as essential	US-07 (Search and Filter Jobs)

Interview Insight	Related User Story
Interviewee suggested a maximum cap on simultaneous applications to prevent indiscriminate bulk-applying, which informed the acceptance criteria for US-04	US-04 (Apply for a Job)
Simple, clearly sectioned interface adaptable to both desktop and mobile was strongly preferred	General system design
From the MO/admin perspective: replacing manual Excel-based screening with a centralised system and fairer workload allocation were identified as primary institutional benefits	US-13 (Review and Process Applications), US-14 (Check TA Workload)

Full interview transcript:

Interviewer: Hi senior, I'm a student from the IOT program, and I'm currently working on the software development project for the TA recruitment system of our university's International School. I'd like to have a quick chat with you. You've had experience applying for and working as a TA, so I'd love to hear your real thoughts and any design suggestions you have for the system. It won't take up too much of your time, thanks so much for your help!

Senior: No problem at all, happy to help. Fire away with your questions.

Interviewer: Great, let's get started then.

Interviewer: Senior, how did you first find out about TA job openings? Through college notices, class groups, or other channels?

Senior: Mainly through the college official website announcements and my major professors posting in our WeChat groups. Occasionally I'd see physical posters in the teaching buildings, but the info was all over the place. Sometimes you'd miss a chance just because you didn't see the notice in time.

Interviewer: Back then, you applied using the school's form and Excel sheet method. Looking back at the whole application process, what was the most troublesome and inconvenient part for you?

Senior: There were quite a few annoying things. First off, you had to fill in all your personal info over and over again for every application—name, major, grades, all that stuff. It got really tedious after a while. Then, once you submitted an application, you'd hear nothing back. You had no idea if the professor had even seen it, or if you'd made it to the shortlist. Just waiting around felt so helpless. Also, the job info was never detailed enough—like the exact workload and working hours. Sometimes you'd start the job and realize it clashed with your schedule.

Interviewer: After you got the TA job, did you ever have trouble communicating with the module organisers or school admins? Like, if you wanted to ask about your work arrangement, was there no easy way to get in touch?

Senior: Yeah, definitely. Sometimes I wanted to ask about specific requirements for grading homework, or switch a proctoring time. The only way was to send an email, and the professors never replied right away. It'd be so much easier if there was a dedicated channel for this kind of communication.

Interviewer: The TA recruitment system we're building is exactly meant to fix these problems. First off, what do you think are the must-have core features for the system?

Senior: I think the most important one is definitely being able to check your application status—just to have a clear idea of where you stand. Then, it'd be great to upload your CV directly, so you don't have to fill in the same info repeatedly, and it's easier for the professors to review too. Also, having all the job openings listed in one place—no more hunting for info everywhere.

Interviewer: Those are all core features we're focusing on building. For browsing job openings, would you want the system to have filter options? Like filtering by course type, working hours, the college offering the job, that sort of thing?

Senior: That'd be so needed! When I was looking for TA jobs, I wanted a teaching assistant role that fit in with my after-class time, but I had to go through every single posting one by one. It took forever. Being able to filter directly would save a ton of time.

Interviewer: Also, do you think a one-click application feature for multiple jobs is necessary? Or would you rather apply for each job carefully one by one?

Senior: A one-click apply feature is fine, but there should definitely be a limit—like a maximum of 2 or 3 jobs at the same time. Otherwise, some people would mindlessly apply for every single opening, hogging all the spots, and the people who actually want the jobs won't get a chance. It'd also make things a lot more work for the school when they're reviewing applications.

Interviewer: That's such a thoughtful point—we'll definitely add that limit to the system. Would you want the system to show the number of applicants for each job, so you can get a sense of how competitive it is? Do you think that's a good thing or not?

Senior: That's a pretty neat feature, I think it's a good addition—especially for first-time TA applicants. It lets you make a more informed call. For example, if a job is super competitive and your qualifications aren't the strongest, you can just go for other openings instead of wasting your time. You don't need to show the exact number though, just a rough range is enough.

Interviewer: Talking about features, let's move on to the system's interface. What's more important to you—simple, easy operation, or a feature-packed, visually appealing interface?

Senior: Simple and easy to use, hands down. We'd use the system for convenience, so if the interface is too complicated and you have to hunt for every function, it defeats the whole purpose. It'd be best if the functions are clearly sectioned off—like a "Personal Center", "Job Browse", "My Applications" tab that you can see at a glance. And it'd be even better if it's mobile-friendly. Sometimes you don't have a computer on hand, so being able to check and apply for jobs on your phone would make it a lot more flexible.

Interviewer: Got it—we'll design the interface to be simple and practical, and make it work on both computers and mobile phones. For system notifications, when you get a reminder for a new TA opening or an update on your application status, would you prefer in-system messages, emails, or SMS?

Senior: In-system messages plus emails are perfect. SMS is just too intrusive—we get so many messages as it is. In-system messages make sure you see it when you log in, and emails act as a backup, so you don't miss any important application updates.

Interviewer: If your application gets rejected, would you want the system to tell you the reason? Like "major mismatch" or "academic performance does not meet the requirements"?

Senior: Yeah, I would. Even just a simple reason is fine. It lets you know where you fall short, so you can improve for future applications, and it stops you from overthinking and guessing why you didn't get it.

Interviewer: These suggestions are incredibly valuable—we'll note them all down and factor them into the system design. One last question: besides making things easier for applicants, what do you think is the biggest value this TA recruitment system can bring to the school's module organisers and administrators?

Senior: For the school side, it should definitely boost recruitment efficiency. They won't have to sort through application info in Excel sheets one by one anymore—the system can compile it all automatically, making it easier to screen applicants. Also, it'll help allocate TA workloads more fairly, so some TAs aren't swamped with work while others have nothing to do. It'll make the school's management a lot more standardized.

Interviewer: Thank you so much, senior! Your experience and suggestions are absolutely crucial for our system design. You've helped us work through a lot of our design questions, and it'll make our system so much more tailored to real user needs.

Senior: Anytime. Hope your system gets built smoothly, and that it makes applying for TA jobs a lot easier for the juniors and sophomores after me.

Interviewer: We'll work our hardest on it! Sorry for taking up so much of your time, and thanks again for your time and help. Wish you all the best for your graduation and everything ahead!Senior: Thank you, and good luck with your project too!

Post-Interview Notes

This interview gathered key insights from a graduating senior with hands-on TA application and working experience, including the core pain points of the current TA recruitment process,

functional requirements for the system, and design suggestions for the interface and user experience. All information collected will serve as supporting material for the design of BUPT International School's TA recruitment system, and will be fully integrated into every stage of the product's functional design and iterative development.